Springboard Virtual Internship 6.0

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Branch: Computer Science and Engineering [Data -Science]

Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

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Insights Report – U.S. Natural Disaster Analysis

Milestone-1: Temporal Analysis

- Over the decades, disaster declarations in the U.S. have steadily increased, reflecting either a rise in actual events or improved reporting mechanisms.
- Seasonal patterns are evident, with certain months consistently recording higher disaster frequencies.
- A 5-year rolling average highlights long-term growth trends, smoothing out short-term fluctuations.
- Sharp spikes in specific years suggest periods of extreme or multiple disasters.
- Year-over-year growth rates vary significantly, confirming that disasters are unevenly distributed across time.

Milestone-2: Geographic Analysis

- The Southern U.S. emerges as the most disaster-prone region, facing repeated vulnerabilities.
- States such as Texas, Florida, and Louisiana consistently appear in high-risk categories.
- Coastal regions, particularly in the southeast, are hotspots for hurricanes.
- Choropleth maps reveal clear geographic clustering, with disasters concentrated in specific zones rather than evenly spread.
- Top affected states show exposure to multiple disaster types, indicating diverse vulnerabilities.

Milestone-3: Incident Type Analysis

- Certain disaster types dominate the dataset, with weather-related events (storms, floods, hurricanes) being the most frequent.
- Regional variations are clear: different states experience different dominant disaster types.
- Treemaps and pie charts illustrate the proportional contribution of each incident type, emphasizing the dominance of severe storms and floods.
- The distribution is uneven, showing specialized disaster patterns across regions.

Milestone-4: State vs Incident Type & Assistance Programs

- States show strong associations with specific disaster types (e.g., hurricanes in Florida, floods in Missouri).
- Heatmaps reveal clusters of high-intensity disaster activity, highlighting vulnerable regions.
- Some states face multiple disaster types simultaneously, increasing their overall risk profile.
- Assistance programs vary by disaster type: certain incidents demand more public assistance, while others affect individuals directly.
- High-frequency disasters consistently receive structured support, showing a strong link between disaster type and resource allocation.

Combined Insights

- Disaster frequency is rising over time, with clear geographic concentration in vulnerable regions.
- Historical patterns reveal predictability: certain states repeatedly face specific disaster types.
- Disaster types, locations, and assistance programs are interconnected, forming a comprehensive picture of vulnerability.
- High-risk states not only face frequent disasters but also diverse incident types, compounding their challenges.

Final Summary

- Disasters are becoming more frequent and regionally concentrated.
- Weather-related disasters dominate the dataset.
- Assistance programs are closely tied to disaster severity and type.
- Data visualization provides clarity, helping policymakers and communities prepare better for future risks.

Deployed Dashboard

The interactive dashboard for this project has been successfully deployed and can be accessed here:
Live Dashboard: https://u-s-natural-disaster-dashboard.vercel.app/